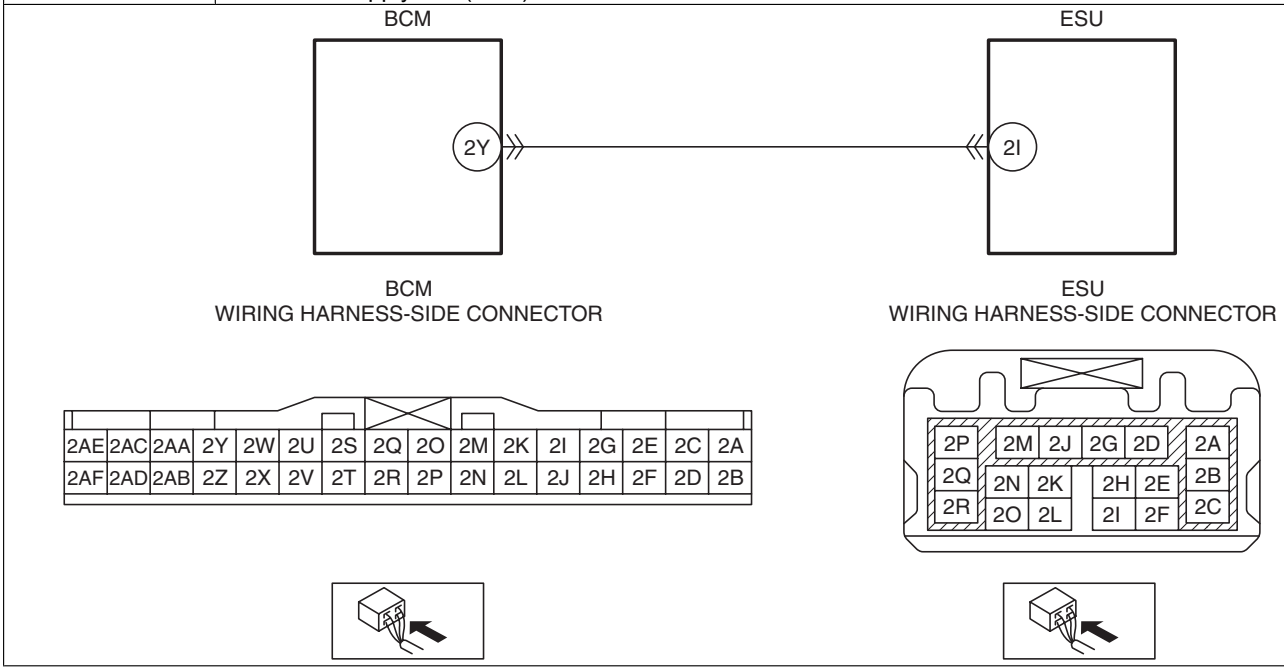
DTC U0422:86 [ELECTRICAL SUPPLY UNIT (ESU)]

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Description	Error signal received from BCM
Detection condition	• Any of the following conditions is met with the ignition switched ON (engine off or on): — The electrical supply unit (ESU) received a headlight LO on request signal error signal from the BCM for **2 s or more**. — The electrical supply unit (ESU) received a headlight HI on request signal error signal from the BCM for **2 s or more**. — The electrical supply unit (ESU) detected a mismatch in the following signal combinations received from the BCM for **2 s or more**. • Headlight variation information signal • Headlight LO on request signal — The electrical supply unit (ESU) detected a mismatch in the following signal combinations received from the BCM for **2 s or more**. • Headlight variation information signal • Headlight HI on request signal
Fail-safe	**Exterior lighting system** • When the ignition is switched ON (engine off or on), the headlights LO, TNS, and panel light turn on and other exterior light controls are stopped.
Possible cause	• DTCs are stored in the BCM. • BCM connector or terminal malfunction • Electrical supply unit (ESU) connector or terminal malfunction • Short to ground in wiring harness between BCM terminal 2Y and electrical supply unit (ESU) terminal 2I • Short to power supply in wiring harness between BCM terminal 2Y and electrical supply unit (ESU) terminal 2I • BCM malfunction • Electrical supply unit (ESU) malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY BCM DTCs** • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [BCM].)
		No	Go to the next step.

Step	Inspection		Action
2	**INSPECT BCM CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the BCM connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.
3	**INSPECT ELECTRICAL SUPPLY UNIT (ESU) CONNECTOR CONDITION** • Disconnect the electrical supply unit (ESU) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.
4	**INSPECT LIN COMMUNICATION CIRCUIT BETWEEN BCM AND ELECTRICAL SUPPLY UNIT (ESU) FOR SHORT TO GROUND** • Verify that the BCM and electrical supply unit (ESU) connectors are disconnected. • Inspect for continuity between BCM terminal 2Y (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between BCM terminal 2Y and electrical supply unit (ESU) terminal 2I. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 7.
		No	Go to the next step.
5	**INSPECT LIN COMMUNICATION CIRCUIT BETWEEN BCM AND ELECTRICAL SUPPLY UNIT (ESU) FOR SHORT TO POWER SUPPLY** • Verify that the BCM and electrical supply unit (ESU) connectors are disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at the BCM terminal 2Y (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between BCM terminal 2Y and electrical supply unit (ESU) terminal 2I. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 7.

Step	Inspection		Action
6	**VERIFY IF MALFUNCTIONING LOCATION IS BCM DEPENDING ON REPEATABILITY** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the electrical supply unit (ESU) using the M-MDS. (See CLEARING DTC [ELECTRICAL SUPPLY UNIT (ESU)].) • Switch the ignition ON (engine off or on) and wait for **2 s or more**. • Retrieve the electrical supply unit (ESU) DTCs using the M-MDS. (See DTC INSPECTION [ELECTRICAL SUPPLY UNIT (ESU)].) • Is the same DTC displayed?	Yes	Replace the BCM, then go to the next step. (See BODY CONTROL MODULE (BCM) REMOVAL/ INSTALLATION.)
		No	Go to Step 8.
7	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the electrical supply unit (ESU) using the M-MDS. (See CLEARING DTC [ELECTRICAL SUPPLY UNIT (ESU)].) • Switch the ignition ON (engine off or on) and wait for **2 s or more**. • Retrieve the electrical supply unit (ESU) DTCs using the M-MDS. (See DTC INSPECTION [ELECTRICAL SUPPLY UNIT (ESU)].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the electrical supply unit (ESU). (See ELECTRICAL SUPPLY UNIT (ESU) REMOVAL/ INSTALLATION.) Go to the next step.
		No	Go to the next step.
8	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [ELECTRICAL SUPPLY UNIT (ESU)].)
		No	DTC troubleshooting completed.